School of Future Tech

Case Study Report

on

ShipShape — Marine Maintenance Services Website

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1. Introduction to the Case Study

ShipShape is a fully functional, multi-page marine maintenance website designed to allow boat owners to book professional vessel services online. The project demonstrates a real-world front-end application built using core web technologies — HTML5, CSS3, and vanilla JavaScript — without any back-end framework.

The site consists of two primary pages: a **Services / Booking page** (*services.html*) and an interactive **Marina Map page** (*map.html*). Together they form a seamless user journey from discovering ShipShape's offerings to locating the nearest marina partner.

This case study explores the architecture, design decisions, site flow, and technical implementation of the ShipShape website, analysing how modern CSS layout techniques, event-driven JavaScript, and third-party mapping libraries combine to deliver a polished user experience for the marine industry.

Sailboat	**Yacht**	**Jet Ski**

2. Problem Statement / Case Background (Abstract)

Background

The marine maintenance industry historically relies on phone calls and in-person scheduling, making it time-consuming and error-prone. Boat owners need a modern digital touchpoint where they can select their vessel type, customise a service package, get a real-time price estimate, and locate a nearby marina — all in a single, intuitive interface.

Abstract

This case study presents the design and implementation of ShipShape, a client-side web application for marine maintenance booking. The site uses a **vessel-aware service filter** to show only relevant services based on the selected boat type, a **live pricing engine** that computes costs from length, service selection, surcharges, and bundle discounts, and an **interactive Leaflet.js map** to display marina locations across South Florida with live technician availability. The project demonstrates best practices in responsive layout, progressive disclosure UI, and modular JavaScript.

3. Case Study Design

ShipShape is architected as a two-page single-responsibility site. Each page has a clear role and the shared stylesheet (*style.css*) ensures visual consistency throughout.

3.1 Page Architecture

services.html	Landing page — hero, about section, booking form
map.html	Marina locator — sidebar list + Leaflet map
style.css	Global styles: nav, hero, cards, sticky bar, map layout
services.js	Booking logic — filtering, pricing, confirmation
marina-map.js	Map initialisation, marker creation, list click handlers

3.2 UI / UX Design Principles

- Progressive disclosure — users are guided through 3 numbered steps.
- Sticky action bar — price and CTA always visible without scrolling.
- Card-based selection — tactile vessel and service cards with active states.
- Responsive grid — collapses gracefully on mobile viewports.
- Semantic colour coding — Ocean blue (#0077B6) for actions, red for emergencies.

4. Methods & Algorithms / Technology Applied

4.1 Vessel-Aware Service Filtering (Algorithm)

Each service card carries a data-vessels attribute listing compatible vessel types (e.g. *sailboat,yacht*). On vessel selection, the JS iterates all service cards and toggles display:block / display:none based on a simple String.includes() check. Unchecked hidden cards are reset to prevent stale pricing.

4.2 Real-Time Pricing Engine

Length base	Boat length (ft) × $4.50 / ft
Service add-on	Base price + $15 (eco) or $85 (emergency) surcharge
Emergency toggle	+ $150 flat fee; label turns red
Monthly bundle	Subtotal × 0.80 (20% discount applied last)

4.3 Leaflet.js Map Integration

The marina map is powered by **Leaflet.js v1.9.4** with OpenStreetMap tiles. A JavaScript array of marina objects (name, lat, lng, technician count) seeds both the marker layer and the sidebar list simultaneously, keeping data in a single source of truth. Clicking a sidebar item calls map.setView() and marker.openPopup(), creating a linked interaction between the list and the map.

4.4 Technology Stack

Structure	HTML5 — semantic markup, data attributes
Styling	CSS3 — Flexbox, Grid, custom properties, sticky positioning
Behaviour	Vanilla JavaScript (ES6) — DOM manipulation, event listeners
Mapping	Leaflet.js v1.9.4 + OpenStreetMap tiles
Fonts	Google Fonts — Outfit (headings), Inter (body)
Dev Env	VS Code / any static file server

5. Implementation Details and Site Flow Diagrams

5.1 Services Page — Step-by-Step Flow

The booking page guides users through three numbered steps before confirming:

User visits services.html

▼

Hero Section displayed — CTA scrolls to booking form

▼

Step 1 — Select Vessel (Sailboat / Yacht / Jet Ski)

▼

filterTheList() called — incompatible services hidden

▼

Step 2 — Enter Boat Length & choose Service Plan

▼

calculateTotal() triggered on every input change

▼

Step 3 — Check desired service cards

▼

Sticky bar updates: price shown, Emergency toggle available

▼

Click 'Confirm Booking' → Confirmation overlay displayed

5.2 Marina Map Page — Flow

User clicks 'Marina Map' in navigation

▼

map.html loads — Leaflet map initialised, centred on South Florida

▼

marinas[] array iterated — 5 markers placed on map

▼

Sidebar list populated with marina name + technician count

▼

User clicks a list item → map.setView() + marker.openPopup()

▼

Active highlight applied to selected list item

5.3 Key Code Snippets

Vessel filter logic (services.js):

```
function filterTheList() { allServiceCards.forEach(card => { const allowed =
card.getAttribute('data-vessels'); card.style.display =
allowed.includes(currentBoat) ? 'block' : 'none'; if
(!allowed.includes(currentBoat)) card.querySelector('input').checked = false; });
}
```

Price calculation summary (services.js):

```
function calculateTotal() { let price = boatLength * PER_FOOT_RATE; // $4.50 / ft
// Add selected service prices + type surcharges allCheckboxes.forEach(box => { if
(box.checked) { price +=
parseFloat(card.querySelector('.price-val').innerText.replace('$','')); if
(box.dataset.type === 'eco') price += 15; if (box.dataset.type === 'emergency')
price += 85; } }); if (emergencyToggle.checked) price += 150; // On-water
surcharge if (planSelect.value === 'monthly') price *= 0.80; // Bundle discount
totalPriceText.innerText = price.toFixed(2); }
```

5.4 Services Offered

Hull Cleaning $75 + $15 eco fee All vessels	Engine Check $90 + $85 surcharge Sailboat & Yacht	Winterization $120 + $15 eco fee Sailboat & Yacht

6. Results and Conclusion

Findings

- The vessel filter successfully shows only compatible services, reducing cognitive load.
- The real-time pricing engine updates instantly, removing the need for a manual 'calculate' button.
- The sticky action bar keeps the price and booking CTA persistently visible, reducing drop-off.
- The Leaflet.js map renders 5 marina markers with interactive popups in under 200 ms on a standard connection.
- The CSS Grid and Flexbox layout collapses cleanly on screens narrower than 600px.
- Separating concerns across three JavaScript files (services.js, marina-map.js) improves maintainability.

Conclusion

ShipShape demonstrates how a well-structured vanilla web stack — HTML5, CSS3, and JavaScript — can deliver a sophisticated, production-quality user experience without any framework overhead. The modular file structure, event-driven pricing logic, and third-party map integration form a solid foundation that could be extended with a backend API, user authentication, and payment processing to create a fully operational marine services platform.

The project exemplifies clean separation of concerns, responsive design, and real-time interactivity — core competencies for any modern front-end developer.

7. References

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